SMAART Career Intelligence Platform – Flow Explanation

1. Overall System Flow

1. Generate Role Data using AI (Claude API)
2. Verify and Store Role Data in Database
3. Student Onboarding
4. Student-Role Matching Engine
5. Role Selection
6. Learning Path Generation
7. Skill Completion Tracking
8. Role Change Handling
9. Skills Passport Generation
10. Engagement Score Calculation

---

2. Detailed Flow Explanation

Phase 1 – Role Data Generation (Build Phase)

During the build stage the system generates detailed data for around 250 career roles.

Flow: Roles List → AI Prompt → Claude API → Structured JSON Data → Narrative Text → Database Storage

Steps: 1. A list of career roles is prepared. 2. For each role the system sends a prompt API. 3. The AI returns structured JSON data containing: - technical skills - AI tools used - priority - salary range - AI exposure 4. Another prompt generates narrative explanations about 5. Developers verify the AI output manually. 6. Verified data is inserted into database tables roles, role_skills, skills and courses.

Important principle: AI is used only during data generation. The running platform does not depe calls for most operations.

Phase 2 – Student Onboarding

When a student first signs up they provide:

• Degree • Specialisation • Career Interest Area • Existing skills

Flow: Student Signup → Enter Degree & Specialisation → Choose Interest Area

The system then prepares to recommend career roles.

Phase 3 – Student to Role Matching

This is done using a prebuilt database table instead of live AI calls.

Flow: Student Input → Query degree_interest_areas → Query degree_role_recommendations → Return 3 Roles

Three roles are recommended:

Primary Role – strongest match Secondary Role – good alternative Alternative Role – different direction

Example: B.Com Finance + Banking Interest → Primary: Financial Analyst Secondary: Credit Analy Alternative: Business Analyst

Because the recommendations are stored in the database the response time is extremely fast.

Phase 4 – Role Selection

Student actions:

Accept recommended role OR Browse full role catalogue

When the student selects a role:

students.primary_role_id is updated primary_role_start_date is recorded

The learning path is then generated.

Phase 5 – Learning Path Engine

The learning path tells the student what skills to learn next.

Flow: Get Role Skills → Get Student Completed Skills → Calculate Missing Skills → Calculate Pric Sort Skills

Priority formula considers:

Role importance Cross role relevance Prerequisite completion

The highest priority skills appear first in the student dashboard.

Phase 6 – Skill Completion Tracking

Students complete skills through:

• Courses • Assessments • Self-report completion

Flow: Student completes skill → Update student_skill_completions table → Learning path recalculates

Coverage example: 5 skills completed out of 15 Coverage = 33%

Phase 7 – Role Change System

Students can change their career role anytime.

Flow: Browse Roles → Compare Skill Coverage → Confirm Switch

When switching:

1. Old role progress is archived
2. New role becomes primary role
3. Learning path recalculates

Database actions:

Insert record in role_archive Update students table

No AI is used here. Only database queries.

Phase 8 – Skills Passport Generation

The Skills Passport is a capability profile for employers.

Eligibility conditions:

• At least 30% skill coverage • At least 3 assessments completed • Minimum 30 days role • English diagnostic passed

Flow: Eligibility Check → Collect Data → Generate Narrative → Create Passport

Only the narrative summary uses AI.

Everything else is generated from database data.

Phase 9 – Engagement Score

A weekly engagement score is calculated for each student.

Formula includes:

Learning progress Assessment completion Recent activity Role stability Course completion

Scores are categorized:

Green – Highly engaged Amber – Moderate engagement Red – Low engagement

This score is visible only to placement officers.

---

3. System Flow Diagram (Conceptual)

Build Phase

Roles List ↓ Claude API Generation ↓ Verification ↓ Database Storage

Runtime Phase

Student Signup ↓ Degree + Specialisation ↓ Interest Area Selection ↓ Role Recommendation Eng
Select Role ↓ Learning Path Generation ↓ Skill Completion ↓ Skill Coverage ↓ Eligibility Check
Passport

---

4. Key Design Principle

The system avoids AI usage during normal platform operations.

AI is used mainly for:

Role data generation Matching table generation Skills passport narrative

Everything else relies on database queries.

This design makes the platform scalable and extremely low cost even for tens of thousands of students.

End of Explanation